MIL-S-24522
23 September 1975

MILITARY SPECIFICATION

SEALER, DECK COVERING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a water-based sealer for use on terrazzo, latex mastic, and magnesite deck coverings.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

MILITARY
MIL-E-1115 - Enamel, White (Formula 30) (For Naval Shipboard Use).
MIL-D-3134 - Deck Covering Materials.
MIL-D-16791- Detergents, General Purpose (Liquid, Nonionic).

STANDARD

FEDERAL
FED-STD-141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling, and Testing.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 92 - Flash and Fire Points by Cleveland Open Cup, Test for.
D 97 - Pour Point of Petroleum Oils, Test for.
D 1391 - Odor in Atmospheres (Dilution Method), Measurement of.
E 70 - pH of Aqueous Solutions with the Glass Electrode, Test for.
G 23 - Operating Light-and-Water-Exposure Apparatus (Carbon-Arc Type) for Exposure of Nonmetallic materials, Rec. Practice for.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The sealer, a water based emulsion of a polymer, shall be a product capable of forming a water and abrasion resistant film.

3.2 Solids. The sealer shall have a minimum of 14.0 percent solids, when tested as specified in 4.4.1.

3.3 Flash point. The sealer shall have no flash point, when tested as specified in 4.4.2.

3.4 Freeze-thaw stability. The sealer shall show no separation or gelling, when tested as specified in 4.4.3.

3.5 Slip resistance. The sealer shall have a minimum slip index reading of 5.0, when tested as specified in 4.4.4.

3.6 Moisture absorption. The test specimen shall absorb not more than 1.70 percent moisture, when tested as specified in 4.4.5.

3.7 Flexibility. The sealer shall not exhibit any cracking, when tested as specified in 4.4.6.

3.8 Resistance to wear. No more than 7.0 milligrams (mg) shall be abraded from the test panel, when tested as specified in 4.4.7.

3.9 Chemical resistance. The sealer shall be resistant to mild acid, mild alkali, synthetic detergent, sodium chloride solution, and alcohol, when tested as specified in 4.4.8.

3.10 Drying time. The sealer shall dry hard in no more than 30 minutes, when tested as specified in 4.4.9.

3.11 Aging (yellowing). Loss of reflectance of the test panel shall not exceed 6.0 percent and sealer shall not be removed during this test nor shall the paint film be dulled, when tested as specified in 4.4.10.

3.12 Condition in container. The material as received shall show no evidence of separation, deterioration, or settling when tested as described in 4.4.11. The sealer shall not corrode a metal container.

3.13 pH. The pH value shall be between 7.0 and 9.0 when tested as described in 4.4.12.

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3.14 Odor. The odor of the sealer immediately upon opening the container and during application shall not be pungent. There shall be no residual odor after 24 hours of air drying when tested as specified in 4.4.13.

3.15 Workmanship. The sealer shall be a homogeneous emulsion, free from insoluble, suspended, or precipitated solids. The sealer shall show no evidence of separation, gelling, or solidification during handling or storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling and inspection. Sampling and inspection shall be performed in accordance with methods 1021 and 1031 of FED-STD-141.

4.2.1 Lot. For purposes of sampling, a lot shall consist of all material formulated under the same conditions of production and offered for delivery at one time.

4.3 Quality conformance inspection. Tests for acceptance of individual lots shall consist of all the tests specified in 4.4. Each sample shall be kept separate and analyzed separately.

4.3.1 Rejection. If a sample of sealer is found to be not in conformance to the requirements of the specification, this shall be cause for rejection of the lot represented by the sample.

4.4 Tests

4.4.1 Solids. Solids (nonvolatile matter) shall be determined by method 4041 of FED-STD-141. Thinning solvent shall be 2 milliliters (ml) of water.

4.4.2 Flash point. Flash point shall be determined by ASTM D 92. Test shall be discontinued if no flash point occurs when sample boils for 30 seconds.

4.4.3 Freeze-thaw stability. Apparatus shall be as described in ASTM D 97. A test jar shall be filled to the mark with sealer. The jar shall be stoppered and placed in a cold box, maintained at 0° ± 5° Fahrenheit (F) for 8 hours. The jar shall be removed from the cold box and kept at room temperature (70° 75°F) for 16 hours. If separation or gelling of the sealer has occurred, test shall be discontinued. Otherwise, repeat the freeze-thaw for a total of 3 cycles.

4.4.4 Slip resistance.

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4.4.4.1 Preparation of terrazzo test specimens. Two specimens, each 6 inches by 6 inches by 1/4 inch, shall be prepared in accordance with the manufacturer's instructions from a deck covering conforming to type I, class 1 of MIL-D-3134. The deck covering shall be trowelled on 1/8-inch steel plates covered with a light coating of grease or oil so that upon drying the specimens will not adhere to the plates. A frame shall be used to give the desired dimensions. After 24 hours, the specimens shall be removed from the frames and steel plates and allowed to cure for at least another 72 hours. The specimens shall be grouted according to manufacturer's instructions and allowed to cure for at least 48 hours. The terrazzo specimens shall be ground smooth with a power sander using No. 60 paper without water, unless the manufacturer specifies another abrasive paper or procedure. Tests using these specimens shall be conducted at a temperature of 70° to 75°F and a relative humidity of 48 to 52 percent. One of the terrazzo specimens shall be used for this test; the other shall be retained for the moisture absorption test of 4.4.5. A cheesecloth applicator shall be dipped into the sample of sealer and the excess squeezed out. A coat of sealer shall be applied to the test specimen and allowed to dry. Three more coats of sealer shall be applied to the specimen in the same way. After the fourth coat, the specimen shall be allowed to dry overnight.

4.4.4.2 Apparatus. A Horizontal Pull Slipmeter[1/] or equal shall be used to determine slip resistance. The Slipmeter, a portable instrument weighing 6 pounds, is supported by three removable, 3/4-inch diameter feet, each fitted with shoe sole leather. The Slipmeter shall be pulled over the specimens at a rate of 4.0 inches per minute by means of a cord wound around a winch driven by a constant speed motor.

4.4.4.3 Test procedure. The Slipmeter shall be centered parallel to two sides of a specimen prepared in 4.4.4.1. The Slipmeter shall be operated and a reading (slip index) taken. The reading shall be repeated for a total of ten times with the Slipmeter in the same position. The specimen shall then be turned 180 degrees so that the pull along the specimen can be measured in the opposite direction. Ten additional Slipmeter readings shall be taken on the repositioned specimen in the manner described above. The total of 20 readings shall be averaged.

4.4.5 Moisture absorption.

4.4.5.1 Preparation of terrazzo test specimens. Three specimens, each 2 inches square, shall be cut from the unused specimen prepared in 4.4.4.1. The edges of the cut specimens shall be ground smooth in the same way that the specimen faces were previously ground. A brush coat of sealer shall be applied to one face and the edges of the specimens. When dry, the specimens shall be turned over and a brush coat of sealer shall be applied as previously and allowed to dry. A total of four brush coats of sealer shall be applied to the specimens in the manner described above. After the fourth coat, the specimens shall be allowed to dry overnight before use.

1/Manufactured by Olsen Medical Products, Inc., 50 Megunko Road, Ashland, Massachusetts 01721.

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4.4.5.2 Test procedure. Each specimen shall be dipped in tap water at room temperature, lightly wiped on all surfaces with a paper towel and weighed to the nearest mg on an analytical balance. Immediately after weighing, the specimens shall be immersed separately in tap water for 24 hours, lightly wiped, and reweighed. The percentage gain in moisture shall be based on the difference between the weight of the specimen after 24 hours immersion and the initial weight.

4.4.6 Flexibility.

4.4.6.1 Preparation of test panel. Tinplate used in this test as well as tests listed in 4.4.7 through 4.4.10 shall conform to that described in method 2012 of FED-STD-141. For all of these tests, temperature shall be 70° to 75°F and relative humidity shall be 48 to 52 percent unless otherwise specified (see 6.2). All test panels shall be degreased with trichloroethylene prior to coating. A brush coat of sealer shall be applied to a 6-inch by 6-inch tinplate panel and allowed to dry for 24 hours. A second coat of sealer shall be brushed over the first coat and allowed to dry for 24 hours.

4.4.6.2 Test procedure. Flexibility shall be determined by method 6221 of FED-STD-141, using a 1/2-inch steel rod as a mandrel.

4.4.7 Resistance to wear.

4.4.7.1 Preparation of test panels. A cheesecloth applicator shall be dipped into the sample of sealer and the excess shall be squeezed out. A coat of sealer shall be applied to three tinplate panels (see 4.4.6.1), each 17 inches by 2-1/2 inches, and allowed to dry. Three more coats of sealer shall be applied to the panels in the same way. After the fourth coat, panels shall be allowed to dry overnight.

4.4.7.2 Apparatus. The apparatus[2/] shall consist essentially of an electric motor mounted on a flat surface and a mechanism through which the motor will impart a reciprocating motion lengthwise across a clamped test panel to a piece of 320 J aluminum oxide cloth held in an abrasion boat. A satisfactory arrangement is shown in figure 1 of method 6141 of FED-STD-141.

4.4.7.3 Test procedure. A test panel, prepared in 4.4.7.1, shall be clamped in the apparatus. The panel shall be abraded for 500 cycles. All abraded loose material shall be collected and weighed to the nearest 0.1 mg. This shall include loose material on panel and machine tray as well as anything that can be brushed off the abrading cloth. The test shall be repeated twice using the other panels. Results of the three tests shall be averaged.

4.4.8 Chemical resistance.

4.4.8.1 Preparation of test panels. Four coats of sealer shall be applied to five tinplate panels (see 4.4.6.1), each 4 inches by 4 inches, in the manner described in 4.4.7.1.

2/A WG-2000 Gardner Straight Line Washability and Abrasion Machine, Model M-105A, obtainable from Gardner Laboratory, Inc., 5521 Landy Lane, Bethesda, Maryland 20014 or equal, is acceptable.

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4.4.8.2 Preparation of test liquids. Five hundred ml of the following shall be placed in separate 1500 ml beakers:

(a) Five percent solution of anhydrous citric acid in distilled water.
(b) Five percent solution of sodium bicarbonate in distilled water.
(c) 0.2 Percent solution of nonionic detergent (type 1 of MIL-B-16791) in distilled water.
(d) Five percent solution of sodium chloride in distilled water.
(e) Isopropyl alcohol (99.9 percent).

4.4.8.3 Test procedure. A test panel prepared in 4.4.8.1 shall be placed in each of the test liquids. Panels will be half immersed in the liquids. After 1 hour, panels shall be removed, rinsed with distilled water and allowed to air dry for 15 minutes. Upon examination, panels shall not exhibit any deleterious effects such as a white film, white spots, removal of any of the coating, or damage to the coating.

4.4.9 Drying time. A thin coat of sealer shall be applied to a 5-inch by 8-inch tinplate panel (see 4.4.6.1), using a draw down blade with a clearance of 0.0015 inch. Note dry-hard time according to method 4061 of FED-STD-141. If sealer forms puddles due to poor levelling, time noted shall be when last puddle has dried.

4.4.10 Aging.

4.4.10.1 Preparation of panels. Three 3-inch by 9-inch tinplate panels (see 4.4.6.1) shall be used. One face of each panel shall be covered with a brush coat of formula 30 white paint in accordance with MIL-E-1115, and allowed to dry for 24 hours. A second coat for formula 30 shall be applied and allowed to dry for 24 hours. Reflectance readings of the painted surface shall be taken using a Gardner Multipurpose Reflectometer or equal. The green filter shall be used. A thin coat of sealer shall be applied to the painted surfaces using a cheesecloth applicator from which the excess sealer has previously been squeezed out. After sealer has dried, a second coat of sealer shall be applied and allowed to dry for 24 hours.

4.4.10.2 Test procedure. The panels shall be placed in a light- and water-exposure apparatus, type D or DH as described in ASTM G 23 for 100 hours. A cam which gives a cycle of 102 minutes of sunlight and 18 minutes of distilled water spray shall be used. Temperature shall be 175° ±10°F. The panels shall be removed from the apparatus, rinsed with distilled water and allowed to air dry. The panels shall be examined for deleterious effects and reflectance readings shall be taken. Loss of reflectance shall be calculated as follows:

$$\text{percent loss in reflectance} = \frac{(A - B) \times 100}{A}$$

Where A = Average initial reflectance of painted panels.
B = Average reflectance of panels after 100 hours in apparatus.

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4.4.11 Condition in container. Pour the sealer as received in a cylindrical glass container or graduate and examine by the use of transmitted light for irregularities as specified in 3.12.

4.4.12 pH value. ASTM E 70 shall be used for determination of the pH of an aqueous solution.

4.4.13 Odor. The material shall be tested for odor in accordance with ASTM D 1391.

4.5 Inspection of preparation for delivery. Inspection of the preservation-packaging, packing, palletized loads, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 General. Unless otherwise specified (see 6.2), sealer shall be furnished in 1- or 5-gallon quantity as specified (see 6.2).

5.1.2 Level A. One- and 5-gallon quantities shall be furnished in cans conforming to type V, class 4 of PPP-C-96. Interior of cans shall be provided with a coating to meet the requirements of 3.13 herein and which shall neither affect nor be affected by the sealer. Exterior plan B coating and side seam striping is required. One-gallon cans shall be oblong and shall be provided with a formed bridge type handle. Five-gallon cans shall be round and shall be provided with a wire handle.

5.1.3 Level C. Sealer in quantity specified (see 5.1.1 and 6.2), shall be packaged in a manner which will afford protection against corrosion (see 3.13), deterioration, leakage and damage from the supply source to the first receiving activity for immediate use. The suppliers' normal packaging procedure may be used when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A, B, and C. Sealer in quantity specified (see 6.2), shall be arranged and packed for the level specified in accordance with the appendix to PPP-C-96. For level A packing, when fiberboard boxes conforming to PPP-B-636 are used, the boxes shall be closed, reinforced and waterproofed in accordance with method V of the appendix to the box specification.

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5.2.2 Palletization. Filled cans, packed as specified, shall be palletized in accordance with the appendix to PPP-C-96.

5.3 Marking. In addition to any special marking required by the contract or order (see 5.3.1, 5.3.2, and 6.2), cans, shipping containers, and palletized loads shall be marked in accordance with the appendix to PPP-C-96.

5.3.1 Instruction label. A label describing manufacturer's instructions for use shall appear prominently on every can of sealer.

5.3.2 Special marking. Each can of sealer shall have the following additional information:

(a) Specification number.
(b) Manufacturer's name and address.
(c) Manufacturer's lot number and date of manufacture.
(d) Contract number.
(e) Shelf life marking.

6. NOTES

6.1 Intended use. The sealer covered by this specification is intended to provide a nonporous film for terrazzo surfaces, latex mastic, or magnesite. This film acts as a barrier against the buildup of moisture, dirt, and grime, and protects the terrazzo against excessive wear and deterioration. This material is not a permanent sealer, but will have to be reapplied periodically.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Temperature and relative humidity if other than specified (see 4.4.6.1).
(c) Level of packaging and packing required (see 5.1, 5.2, and 5.2.1).
(d) Unit quantity required (see 5.1.1 and 5.1.3).
(e) Special marking required (see 5.3.2).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Custodians:
Army - MR
Navy - SH
Air Force - 84

Preparing activity:
Navy - SH
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